# Root Cause Analysis of Gaana

## Project Brief :

- Imagine you are a Product Manager for Gaana - a music streaming app based in India.
- The App Store rating in India has dropped from 4.5 to 4 stars in the last 6 months.
- Find out the possible root cause(s) of the problem.

## Assumptions :

- Assume that this happened only on iPhone devices.
- Please list any other assumptions you have made during your analysis.
- The dropping of ratings is happening across India.

## Analyzing the problem :

The Gana rating in India has dropped to 4 stars from 4.5 over the past 6 months, indicating potential user dissatisfaction. Various factors may contribute to this decline, which we can analyze across the following areas.

- AppStore Ratings & Comments Analysis
- User Journey analysis.
- Internal Factors.
- External Factors.

## AppStore Ratings & Comments Analysis :

I have analyzed the comments and ratings from the last 6 months to assess the situation. Users have encountered several issues, which they have detailed in their feedback. Below, we delve into these observations in detail.

- Total number of reviews(In last 6 months) : 121
- 5 star ratings : 10
- 4 star ratings : 3

- 3 star ratings : 5
- 2 star ratings : 3
- 1 star ratings : 100

**We can find the detailed analysis on ratings and comments in below table along with metrics.**

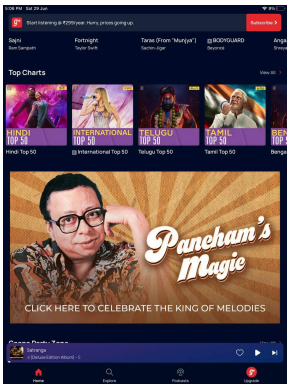
| Theme | Summary of the issue | No of Reviews | Review (%) |
|---|---|---|---|
| Subscription issue. | Subscribed users are encountering numerous challenges while using the application, leading them to perceive it as a scam or fraud due to its significant failure to meet their expectations. | 50 | 41.32 |
| Application issue | The application crashes during song playback, with users reporting numerous bugs. | 26 | 21.48 |
| Bad collection of songs. | The Gaana library lacks diverse and satisfactory song collections, leaving users highly disappointed. | 25 | 20.66 |
| Suggestions issue | Users are receiving incorrect suggestions, such as receiving few devotional songs after playing the top 50 Bollywood Hindi songs, despite not preferring that genre. | 6 | 4.95 |
| Advertising issue | Users, including some subscribers, are experiencing | 4 | 3.30 |

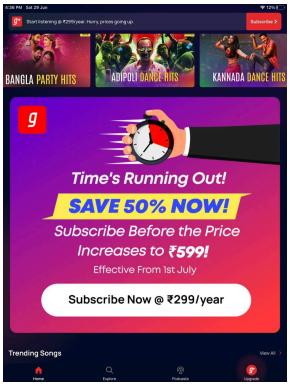
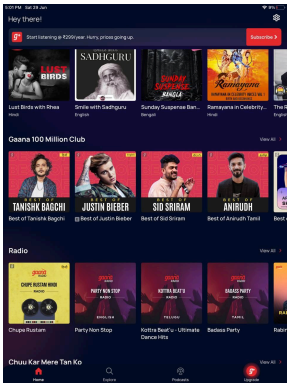
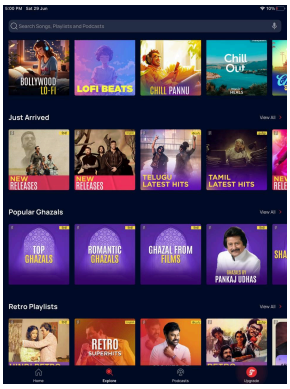
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|---|---|---|---|
| | significant irritation due to frequent advertisements while using the application. | | |
| Issue with bad Customer support | Users' experiences with customer support were not satisfactory. | 3 | 2.47 |
| Search bar issue | Users' are facing difficulty in finding songs using the search bar. | 2 | 1.65 |
| Song Downloading issue | Despite subscribing to the application, users are unable to download certain songs. | 2 | 1.65 |
| Unable to Cast | Users are unable to cast songs to their other devices. | 1 | 0.82 |
| Multiple device issue | The subscribed user is unable to login to other devices. | 1 | 0.82 |
| Song playback doesn't resume after call ends | After ending a call, the user is unable to resume the song and has to replay it from the beginning. | 1 | 0.82 |

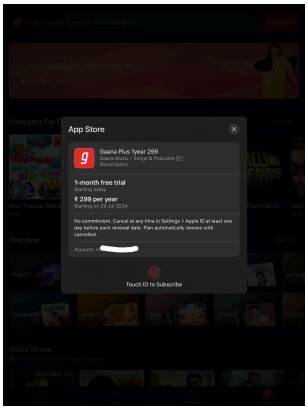
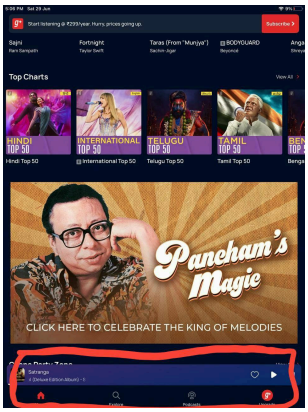
**Based on the data from the app store, the average rating from 121 reviews is only 1.51 stars.** This indicates that the application has several issues affecting users. Let's delve into the user journey analysis to gain more insights into the user perspective.

## User Journey Analysis :

User journey analysis is a fundamental process that examines the user's perspective. The challenges and difficulties encountered by users during their journey serve as key inputs for our analysis. Let's examine this in detail.

| S.No | User Journey | Apple device Screen | Potential problems | Metric to track |
|---|---|---|---|---|
| 1 | Open App |  | It takes time to open the application. | • Time taken to load the home page. |
| 2 | Home Page |  | It appears cluttered and somewhat disorganized. | • Average time spent on the home page. |

| | | | | |
|---|---|---|---|---|
| 3 | Home Page |  | A large advertisement banner on the home screen makes it look cluttered. | • Average time spent on the home page. |
| 4 | Personal recommendations |  | The personal recommendations for me are inadequate and irrelevant. | • Average time spent on this page.<br>• Number of selections made after spending time. |
| 5 | New arrivals or recent songs added |  | The user has to scroll deep to access recently added songs. | • Average time spent on this page |

| | | | | |
|---|---|---|---|---|
| 6 | Subscription |  | Auto-renewal subscriptions often lead users to hesitate, as there's a high risk of forgetting the renewal date, especially if they're dissatisfied with the application. | • Number of subscriptions taken per week/month. |
| 7 | Issue with playback or removing a song from the home screen. |  | The user is having trouble removing a song from the home screen that they played earlier. | • Average time spent on this page. |

Based on the data above, while some points are valid, user journey analysis highlights subscription issues and suggestions as critical areas requiring serious consideration. Given significant negative feedback on subscriptions, thorough analysis with robust data and statistics is essential to improve the user journey in this area.

## Internal Factors :

1. **Technical issues and bugs:**
    - **Hypothesis** : Recent updates may have introduced technical issues in the application, resulting in dissatisfaction among customers.

- Based on my analysis, approximately **21.48%** of reviews indicate the presence of bugs and technical issues in the application. It is crucial to address these bugs promptly by prioritizing communication with the engineering team.
- **This Hypothesis is valid.**

2. **Subscription issue:**
    - **Hypothesis:** We hypothesize that a significant number of subscribers are dissatisfied with our services and feel their expectations have not been met.
    - **Services Available for Subscribed Users:** Based on my analysis, **41.32%** of users who are subscribed express dissatisfaction. They report challenges in accessing the privileges that Gaana offers to subscribed users and express unhappiness about the unavailability of these services to them.
    - This **Hypothesis** is **valid**.

3. **Payment Issue :**
    - **Hypothesis:** We hypothesize that there is a payment gateway issue preventing users from using the application.
    - Based on my review analysis, none of the users have reported experiencing payment issues.
    - **Therefore, this hypothesis is invalid.**

4. **Media library issue :**

    - **Hypothesis**: We believe that over the last 6 months, the media library has been poorly maintained and the algorithm for recommending suggestions to users is not performing well.
        A. **Limited content and poor collection:** In my recent analysis, **20.66%** of users have expressed dissatisfaction with the media library. For example, a user gave a **1-star rating** because they couldn't find the song **"Om Namo Venkatesaya"**
        B. **Irrelevant suggestions :** Based on my analysis , **4.95%** of users have received irrelevant suggestions.

- . Therefore, **this Hypothesis is valid.**

**5. UI/UX Design :**

- **Hypothesis:** We hypothesize that the UI/UX is poor and appears cluttered for users, leading to dissatisfaction.
- Upon review analysis , none of the users have mentioned any issues with the UI/UX.
- **Therefore, this hypothesis is invalid.**

**6. Underperformance of Engineering team:**

- **Hypothesis:** We believe that inefficiencies in our development and QA engineering teams over the last two quarters have led to numerous defects and bugs in the application.
- Based on my review analysis, 4.95% of users have reported receiving irrelevant suggestions and recommendations, and 21.4% of user reviews mention bugs and defects in the application.
- **Therefore, this hypothesis is valid.**

## External Factors :

1. **Government Policy changes/Regulations:**
    - **Hypothesis:** We hypothesized that recent government policy changes might have made users unhappy with music streaming applications.
    - Based on my review, no such regulations or policies have been observed that could lead to user dissatisfaction with music streaming applications.
    - **This hypothesis is invalid.**

2. **Competitors on market :**
    - **Hypothesis:** We believe that competitors offering superior services may affect user satisfaction.

- According to recent surveys, Spotify holds a significant market share in music streaming. JioSaavn and Wynk are also strong players in the marketplace. While this hypothesis may not directly explain the drop in ratings, it suggests areas where Gaana can improve to better compete in the market.
- **This hypothesis is Invalid.**

3. **IOS issues:**
    - **Hypothesis:** We hypothesize that there may be issues with the existing iOS platform affecting our application's performance.
    - Based on my review analysis , **21.2%** of user reviews mention application bugs and defects. There is a possibility that iOS may not run our application as effectively as the Android OS does.
    - **Further investigation into OS performance could validate this hypothesis.**

4. **Recent surges in TROI tariffs:**
    - **Hypothesis:** We hypothesize that recent increases in mobile internet prices are making our users unhappy.
    - Based on my review analysis, there is no evidence suggesting that increases in data prices have resulted in lower ratings in the app store.
    - **Therefore, this hypothesis is invalid.**

5. **Natural calamities and Internal emergencies:**
    - **Hypothesis:** We hypothesize that natural calamities and internal emergencies leading to government-mandated internet shutdowns (ex, in Jammu and Kashmir and Manipur) may cause a drop in application ratings.
    - Based on my recent analysis, there are no user reviews mentioning such issues. Additionally, these events typically affect all internet-dependent services like WhatsApp, Facebook, Instagram, etc., not just our application.
    - **Therefore, this hypothesis is invalid.**

6. **Human Psychology:**
    - **Hypothesis:** We believe that GenZ users who subscribe to our service are predominantly interested in pop and rap songs, which are limited in our application, leading to dissatisfaction.
    - Based on my review analysis, **20.66%** of user reviews mention dissatisfaction with the song collection. Users who prefer pop, hip-hop, and rap may not find their desired songs in our library, which could contribute to lower ratings.
    - **Therefore, this hypothesis is valid.**

## Metrics :

- **North star Metric :** Total number of users with subscription.
- L2 Metric : Privileges which are available for the subscribed users. (No of songs, No adds, No banners etc)

## Conclusions:

- **The user journey does not appear to be a significant factor contributing to the decline in ratings.**
- **Internal factors such as subscription issues, technical issues, and a limited media library are major contributors to the decline in ratings.**
- **External factors such as human psychology and issues related to iOS may also play a role in the decline in ratings.**